

A · Data harmonization

CHIRPS v2.0

precipitation · 0.05°

ERA5 monthly

t2m · tp · d2m $\rightarrow$ q

MODIS LST

MOD/MYD11C3

Harmonized grid

0.05° · monthly · 2000–2023
200 × 600 × 288 · 8 variables

Study region & patches

SE USA · 3 subregions, 32 × 32 px

B · Sparse discovery

Candidate library Θ

seasonal $\sin/\cos(2\pi t/12) \cdot T, q, P$
spatial derivatives ($\partial x, \partial y, \partial^2 x, \partial^2 y$)
interactions: $T \cdot q, T \cdot q_y, q \cdot q_y$

STLSQ sparse regression

OLS + sequential thresholding
refit on support

Discovered tendency equations

$\partial T / \partial t = a \cdot \sin(2\pi t/12) + b \cdot \partial T / \partial y$
(+ coupled $T \cdot q$ terms)

Weak-form cross-check (WSINDy)

seasonal term as exogenous forcing
· integrated test functions
(PySINDy, WeakPDELibrary)

C · Validation

Spatial cross-validation

hold-out training on each subregion

Block bootstrap

selection frequency & CIs (100 draws)

Forward-integration skill

vs. persistence · climatology ·
fit-once ARIMA

Resolution sensitivity

0.05° vs 0.25°

Per-pixel coefficient maps

spatial structure across domain